Doping Stability Report -- Ni in KCoO2 at 300 C

Generated: 2026-05-30 16:10   Host: KCoO2   Dopant: Ni

Executive Summary

Ni preferentially occupies the Co layer. The formation energy difference between the two layers is +5.907 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is -0.01023 A^2/ps (stable), compared to 0.00762 A^2/ps on the K site. The Co-site result is isovalent (no charge mismatch), so the formation energy is fully reliable. The K-site E_f is approximate due to a charge surplus of +2 that cannot be modelled in CHGNet.

Site	E_f (eV)	Charge	Compensation	MSD pass	Coord pass	Verdict
Co	+0.591	+0	No	PASS	FAIL	STRUCTURAL COLLAPSE
K	+6.498	+2	Yes	FAIL	FAIL	UNSTABLE (migration + structu

Ni at the Co layer -- STRUCTURAL COLLAPSE

Charge situation

Ni replaces Co with no change in oxidation state (isovalent substitution, charge mismatch = 0). No charge compensation is needed. The formation energy is fully reliable.

Formation Energy (thermodynamic stability)

Formation energy E_f	**+0.5915 eV**	PASS	*< 1.0 eV to pass*

E_f = +0.591 eV. This value is moderately positive, indicating that Ni incorporation is possible but endothermic; may require non-equilibrium synthesis at the Co site.

Molecular Dynamics Stability (300 C, 25 ps)

MSD slope (late-time)	**-0.01023 A^2/ps**	PASS	*< 0.005 to pass (plateau = stable)*
Final MSD	**0.518 A^2**	----	
Max displacement	**1.192 A**	----	
Coordination (min/mean/max)	**3 / 4.7 / 5**	FAIL	*relaxed = 5, tolerance +/-1*
Mean bond length (MD vs relaxed)	**1.997 A (relaxed: 2.(**	----	
Lattice volume change	**+0.00 %**	PASS	*< +/-5% to pass*

Late-time MSD slope = -0.01023 A^2/ps. The negative slope is a flat plateau -- essentially no migration detected. Final MSD = 0.518 A^2; maximum displacement from starting position = 1.19 A.

Coordination fluctuated between 3 and 5 (mean 4.7), deviating by more than +/-1 from the relaxed value (5). This may indicate structural distortion, or that the bond-length cutoff is too close to the actual bond length -- check the coordination_cutoff_A setting. Mean nearest-neighbour distance during MD = 1.997 A.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: STRUCTURAL COLLAPSE

The coordination criterion failed while MSD and volume both pass, suggesting a possible cutoff artifact rather than a true structural collapse. The Ni atom barely moves (MSD = 0.518 A^2) but the coordination count fluctuates because the bond length at the Co site is close to the analysis cutoff. Consider increasing coordination_cutoff_A and re-running analysis.

Ni at the K layer -- UNSTABLE (migration + structural distortion)

Charge situation

Ni replaces K with a charge surplus of +2. CHGNet cannot model this compensation (would require Co oxidation or electron addition). The formation energy below is approximate.

Formation Energy (thermodynamic stability)

Formation energy E_f	**+6.4984 eV**	FAIL	*< 1.0 eV to pass*

E_f = +6.498 eV (approximate -- charge surplus not modelled in CHGNet). This value is large and positive, indicating that Ni incorporation is thermodynamically unfavorable under equilibrium conditions at the K site.

Molecular Dynamics Stability (300 C, 25 ps)

MSD slope (late-time)	**0.00762 A^2/ps**	FAIL	*< 0.005 to pass (plateau = stable)*
Final MSD	**1.251 A^2**	----	
Max displacement	**1.190 A**	----	
Coordination (min/mean/max)	**3 / 4.2 / 7**	FAIL	*relaxed = 4, tolerance +/-1*
Mean bond length (MD vs relaxed)	**1.887 A (relaxed: 1.8**	----	
Lattice volume change	**+0.00 %**	PASS	*< +/-5% to pass*

Late-time MSD slope = 0.00762 A^2/ps. The slope exceeds the plateau threshold, suggesting mild drift or site-hopping. Final MSD = 1.251 A^2; maximum displacement from starting position = 1.19 A.

Coordination fluctuated between 3 and 7 (mean 4.2), deviating by more than +/-1 from the relaxed value (4). This may indicate structural distortion, or that the bond-length cutoff is too close to the actual bond length -- check the coordination_cutoff_A setting. Mean nearest-neighbour distance during MD = 1.887 A.

Volume change = +0.00%. The host lattice is mechanically intact.

VERDICT: UNSTABLE (migration + structural distortion)

Verdict: UNSTABLE (migration + structural distortion). Refer to the individual criteria above.

Site Preference Comparison

The Co site is preferred by 5.907 eV over the K site. Formation energy: +0.591 eV (Co) vs +6.498 eV (K).

Ni preferentially occupies the Co layer. The formation energy difference between the two layers is +5.907 eV, strongly favouring the Co site. MD confirms this preference: on the Co site the dopant MSD plateau slope is -0.01023 A^2/ps (stable), compared to 0.00762 A^2/ps on the K site. The Co-site result is isovalent (no charge mismatch), so the formation energy is fully reliable. The K-site E_f is approximate due to a charge surplus of +2 that cannot be modelled in CHGNet.

Caveats and Limitations

? **CHGNet is charge-neutral.**

The potential does not model oxidation states, polaron formation, or electron/hole localisation. Formation energies for aliovalent dopants (charge mismatch != 0) are approximate even when structural compensation defects are included.

? **No Freysoldt/Kumagai correction.**

Electrostatic finite-size corrections for charged defects require the host dielectric tensor and the DFT electrostatic potential, neither of which is available from a neural-network potential. For publication-quality E_f of charged defects, follow up with DFT using doped or pymatgen-analysis-defects.

? **Metal-rich reference state.**

Chemical potentials are computed from elemental ground-state structures. Synthesis under oxide-rich or other conditions shifts all E_f values by additive constants; the relative ordering of sites is unaffected.

? **MD timescales are short (25 ps default).**

This is sufficient to assess local site stability and rapid migration, but cannot resolve slow diffusion mechanisms. A non-plateauing MSD confirms instability; a plateau does not guarantee stability on longer timescales.

? **Energy rankings are more trustworthy than absolute values.**

CHGNet is a universal MLIP trained on r2SCAN DFT. Formation-energy errors for transition-metal oxides are typically 50-150 meV/atom. Use relative orderings (which site is preferred?) as the primary result.